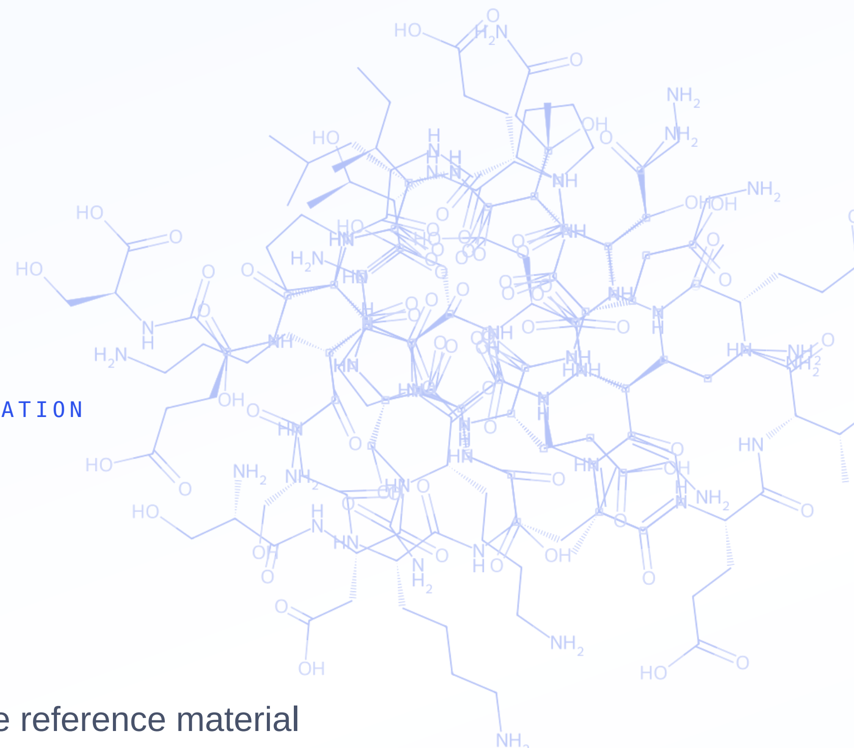
ANVIL COMPOUNDS

RESEARCH USE ONLY

● RESEARCH GRADE · 99%+ PURITY · USA-BASED



RESEARCH REFERENCE & SAFETY DOCUMENTATION

TB-500

Thymosin β4 fragment · Research-grade reference material

PARENT	ACTIVE DOMAIN	CLASS	FORM
Thymosin β4 (43 aa)	**LKKTETQ**	**Peptide fragment**	**Lyophilized**

Identity & Composition

TB-500 is synthetic material based on the actin-binding region of thymosin β4 (Tβ4), a 43-amino-acid actin-regulating peptide found in most human cells. Supplied lyophilized, research grade.

Product name	TB-500
Parent molecule	Thymosin β4 (Tβ4), 43 aa
Parent CAS / MW	77591-33-4 / ≈4921 g/mol
Active domain	LKKTETQ (Tβ4 ~17–23)
Fragment CAS / formula / MW	Depends on product supplied
Appearance	White lyophilized powder
Solubility	Soluble in water
Identified use	In-vitro research & development only

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SDKPDMAEIEKFDKSKLKKTETQEKNPLPSKE
TIEQEKQAGES
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Thymosin β4 sequence (43 aa) with the actin-binding domain (LKKTETQ) highlighted. TB-500 corresponds to synthetic material based on this active region.

> **Identity note.** In this category "TB-500" is applied inconsistently to both full-length thymosin β4 and a shorter synthetic fragment, and most primary research studies full-length Tβ4. Confirm which your supplier ships and reconcile the CAS / formula / MW accordingly before release.

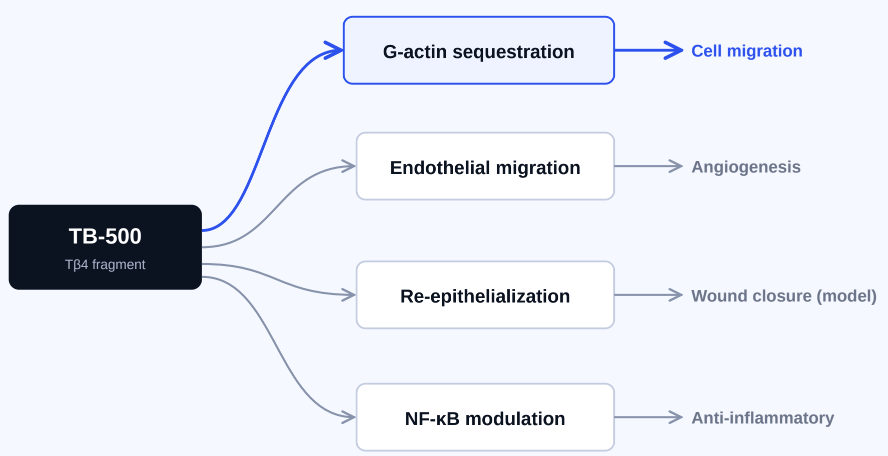
Research Background

Thymosin β4 is the principal intracellular G-actin–sequestering molecule in eukaryotic cells, regulating the actin cytoskeleton that governs cell shape, motility and migration. TB-500 (its active region) has been studied in preclinical tissue-repair research across wound-healing, cardiovascular, corneal and musculoskeletal models since the late 1990s–2000s.

Mechanisms studied in preclinical models

Findings below derive from animal and cell-culture systems. They are not established effects in humans.

- **Actin sequestration (foundational).** High-affinity 1:1 binding to monomeric G-actin (reported Kd ≈ 0.5–0.7 µM), influencing polymerization and cell migration.
- **Cell migration & re-epithelialization.** Promotes keratinocyte and fibroblast migration in preclinical wound models.
- **Angiogenesis.** Endothelial migration and tubule formation; the LKKTET motif is implicated (mutation reduces activity).
- **Anti-inflammatory modulation.** NF-κB-associated modulation of inflammatory signaling in injury models.



Proposed pathways investigated in preclinical (animal / in-vitro) models. Schematic only; not a claim of established human effect.

03 STATE OF THE EVIDENCE

What the Literature Supports — and What It Doesn’t

The actin-sequestration mechanism and preclinical wound-healing / angiogenesis findings are documented across multiple laboratories. A key nuance: most primary evidence concerns full-length thymosin β4, not the TB-500 fragment as sold by reagent vendors.

Where the evidence sits

Full-length Tβ4 (majority of primary literature) Direct TB-500 fragment studies (limited; MDPI 2026 scoping review)

- Most robust evidence concerns **full-length Tβ4**; sequence and quality vary between vendor sources of “TB-500.”
- The legitimate clinical path is **RGN-259** (a Tβ4-based ophthalmic candidate) — as of 2026 at phase 3, with **no approved marketed product**.
- **No human randomized controlled trial** supports systemic use for musculoskeletal injury; no regulatory approvals for therapeutic use.

> **Appropriate characterization:** a well-studied actin-regulating peptide with substantial preclinical tissue-repair literature and one active pharmaceutical program (ophthalmology), awaiting approved clinical outcomes.

Regulatory & status notes

US FDA	No approved TB-500 / thymosin β4 therapeutic product. Certain compounded peptides in this class flagged over limited human safety data.
WADA	Thymosin β4 / TB-500 included on the World Anti-Doping Agency Prohibited List.
Supply basis	Research reagent for in-vitro laboratory use only; not for human or veterinary administration.

Selected references

1. Thymosin Beta-4 and TB-500 in Tissue Healing, Regeneration, and Musculoskeletal Repair: A Scoping Review. Applied Sciences (MDPI). 2026;16(12):6202.
2. Goldstein AL, Kleinman HK. Thymosin β4 in tissue repair and wound healing (foundational literature).
3. Bock-Marquette I, et al. Thymosin β4 mediates cardioprotective effects (porcine MI model).
4. Malinda KM, et al. Thymosin β4 accelerates wound healing. J Invest Dermatol. 1999.

Safety Data Sheet

Prepared per GHS Rev. 8 / OSHA HazCom 2012
(29 CFR 1910.1200) · 16-section format

Note. This Safety Data Sheet is compiled from public sources and weight-of-evidence assessment. Independent review by a qualified chemical-safety professional is recommended prior to use. Supplier identification and emergency-contact details must be completed by the supplier.

1 Identification

Product	TB-500 (thymosin β4 fragment)
Parent CAS	77591-33-4 (full-length Tβ4)
Fragment CAS / formula / MW	Depends on product supplied — confirm
Use	In-vitro research only
Restriction	Not for human/veterinary, food, drug, cosmetic, clinical, or therapeutic use
Supplier	
Emergency	

2 Hazard Identification

Not classified based on currently available data. Limited toxicological data; absence of classification is not a determination of absence of hazard. **Signal word:** None. **Pictograms:** None required. **Precautionary:** P261, P264, P280, P501. Handle as a potentially bioactive peptide of unknown toxicity.

3 Composition

Single-substance product, >98% research grade. Residual synthesis reagents, solvents and counter-ions may be present consistent with research-grade material. See lot COA for impurity profile.

4 First-Aid Measures

Eyes: rinse with water. **Skin:** wash with soap and water; remove contaminated clothing. **Inhalation:** fresh air. **Ingestion:** rinse mouth; do not induce vomiting unless directed. Treat symptomatically.

5 Fire-Fighting

Media: CO_2, dry chemical, foam, water spray. Combustion may produce CO, CO_2, NOx. Use SCBA and full protective gear.

6 Accidental Release

Avoid dust; ventilate; use PPE per §8. Prevent entry to drains/waterways. Collect in a closed labeled container for disposal per §13.

7 Handling & Storage

Handle as bioactive peptide of unknown toxicity; minimize exposure. Weigh/reconstitute in fume hood. Store lyophilized solid tightly closed, protected from light/moisture/heat; −20 °C long-term, 2–8 °C short-term. Aliquot reconstituted solutions; store frozen. Observe COA retest date.

8 Exposure Controls / PPE

No established OEL; control to ALARA. Fume hood for dry powder. Nitrile gloves (double-glove for neat powder); ANSI Z87.1 eye protection; lab coat; N100/P100 respirator where controls insufficient.

9 Physical & Chemical Properties

State	Lyophilized powder
Appearance	White powder
Odor	Odorless
Solubility	Water
MP/BP/Flash	Not determined

10 Stability & Reactivity

Stable under normal storage. Avoid heat, moisture, sunlight, UV, open flame. Incompatible with strong oxidizers, strong acids/bases, strong reducing agents. Decomposition: CO, CO_2, NOx. No hazardous polymerization.

11 Toxicological Information

Not fully characterized. No LD_{50}/LC_{50} available. Skin/eye: treat as potentially irritating. Sensitization: cannot be ruled out. Mutagenicity, carcinogenicity, reproductive toxicity, STOT, aspiration: no substance-specific data; not classified based on available data; hazards cannot be excluded. Avoid exposure if pregnant, nursing, or planning pregnancy.

12 Ecological Information

No substance-specific ecotoxicity, persistence or bioaccumulation data identified. Avoid release to surface water, soil, drains or sewers.

13 Disposal

Dispose per applicable regulations; US per 40 CFR 261–270 (RCRA). Do not dispose into sewers/waterways. Use a licensed waste disposal company.

14 Transport Information

Not regulated as dangerous goods under DOT/IATA/IMDG based on current classification. UN number: not applicable. Shipper must independently verify.

15 Regulatory Information

US TSCA: may qualify for R&D exemption (40 CFR 720.36). OSHA HazCom 2012 aligned with GHS Rev. 8. EU REACH: supplied for SR&D use; verify exemption. Not a legal determination of regulatory status.

16 Other Information

Revision [DATE] · Version 1.0. Independent professional review recommended prior to use. Sources: PubChem; peer-reviewed literature; OSHA HazCom 2012; GHS Rev. 8; OECD Test Guidelines. Provided without warranty. For scientific R&D use only.